

# Bit Manipulation Interview Cheat Sheet (Easy → Medium)

This is the condensed version of everything you need for most interview and OA bit-manipulation problems.

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## 1. Binary Basics

### Powers of 2

$$2^0 = 1$$

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 32$$

$$2^6 = 64$$

$$2^7 = 128$$

$$2^8 = 256$$

$$2^9 = 512$$

$$2^{10} = 1024$$

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### Binary Place Values

Bit Position

3 2 1 0

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8 4 2 1

Example:

1101

$$= 8 + 4 + 1$$

= 13

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## 2. Operators

### AND (&)

$1 \& 1 = 1$

$1 \& 0 = 0$

$0 \& 1 = 0$

$0 \& 0 = 0$

Meaning:

Both bits must be 1.

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### OR (|)

$1 | 1 = 1$

$1 | 0 = 1$

$0 | 1 = 1$

$0 | 0 = 0$

Meaning:

At least one bit is 1.

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### XOR (^)

$1 \wedge 1 = 0$

$1 \wedge 0 = 1$

$0 \wedge 1 = 1$

$0 \wedge 0 = 0$

Meaning:

Bits are different.

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## NOT (~)

$$\sim n = -(n + 1)$$

Examples:

$$\sim 5 = -6$$

$$\sim 10 = -11$$

$$\sim 0 = -1$$

---

## Left Shift

$$n \ll k$$

Formula:

$$n * (2^k)$$

Examples:

$$5 \ll 1 = 10$$

$$5 \ll 2 = 20$$

---

## Right Shift

$$n \gg k$$

Formula:

$$\text{floor}(n / 2^k)$$

Examples:

$$20 \gg 1 = 10$$

$$20 \gg 2 = 5$$

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## 3. Most Important Identity

$$1 \ll k$$

Creates a number having only kth bit set.

Examples:

$1 \ll 0 = 0001$

$1 \ll 1 = 0010$

$1 \ll 2 = 0100$

$1 \ll 3 = 1000$

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## 4. Check / Set / Clear / Toggle

### Check kth Bit

$(n \& (1 \ll k)) \neq 0$

Alternative:

$((n \gg k) \& 1) == 1$

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### Set kth Bit

$n \mid (1 \ll k)$

---

### Clear kth Bit

$n \& \sim(1 \ll k)$

---

### Toggle kth Bit

$n \wedge (1 \ll k)$

---

## 5. Odd / Even

### Check Odd

$(n \& 1) == 1$

---

## Check Even

`(n & 1) == 0`

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## 6. Power of Two

### Observation

A power of 2 has exactly one set bit.

1 = 0001  
2 = 0010  
4 = 0100  
8 = 1000  
16 = 10000

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### Check Power of Two

`n > 0 && (n & (n - 1)) == 0`

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## 7. Famous Bit Tricks

### Remove Rightmost Set Bit

`n & (n - 1)`

Example:

1100

↓

1000

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### Extract Rightmost Set Bit

`n & (-n)`

Examples:

$$12 \& (-12) = 4$$

$$10 \& (-10) = 2$$

$$40 \& (-40) = 8$$

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## Two's Complement

$$-n = (\sim n) + 1$$

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## 8. Count Set Bits

### Brian Kernighan

```
int count = 0;
```

```
while(n != 0){  
    n &= (n - 1);  
    count++;  
}
```

Complexity:

$O(\text{number of set bits})$

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### Built-in

`Integer.bitCount(n)`

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## 9. XOR Mastery

### Golden Properties

$$a \oplus a = 0$$

$$a \oplus 0 = a$$

$$a \oplus b \oplus a = b$$

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## **XOR is Commutative**

$$a \oplus b = b \oplus a$$

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## **XOR is Associative**

$$(a \oplus b) \oplus c$$

=

$$a \oplus (b \oplus c)$$

---

# **10. XOR Problem Recognition**

## **Every Element Appears Twice Except One**

Use:

```
ans ^= nums[i];
```

Problem:

Single Number

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## **Missing Number**

XOR indices and values.

Problem:

Missing Number

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## **Hamming Distance**

$x \wedge y$

Then count set bits.

Problem:

Hamming Distance

---

## Two Unique Numbers

Use:

xor = all numbers XOR

diffBit = xor & (-xor)

Problem:

Single Number III

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## One Number Appears Once, Others Thrice

Count bits.

Use:

count % 3

Problem:

Single Number II

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# 11. Number Complement Pattern

Problem:

Number Complement



Formula:

$(\sim \text{num}) \& \text{mask}$

Mask:

```
int mask = 1;
```

```
while(mask < num)
    mask = (mask << 1) | 1;
```

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## 12. Bitmasking Basics

### Represent Set

0000 -> {}

0001 -> {0}

0011 -> {0,1}

1111 -> {0,1,2,3}

---

### Add Element

```
mask |= (1 << k);
```

---

### Remove Element

```
mask &= ~(1 << k);
```

---

### Check Element

```
(mask & (1 << k)) != 0
```

---

### Toggle Element

```
mask ^= (1 << k);
```

---

## 13. Generate All Subsets

### Core Loop

```
for(int mask = 0; mask < (1 << n); mask++)
```

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### Check Included Elements

```
for(int i = 0; i < n; i++){  
  
    if((mask & (1 << i)) != 0){  
  
        // nums[i] belongs to subset  
    }  
}
```

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### Complexity

$O(n \cdot 2^n)$  $O(n \cdot 2^n)$  $O(n \cdot 2^n)$

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## 14. Pattern Recognition Table

| Problem Clue     | Pattern         |
|------------------|-----------------|
| Odd / Even       | $n \& 1$        |
| Power of 2       | $n \& (n-1)$    |
| Count set bits   | Brian Kernighan |
| Unique element   | XOR             |
| Missing number   | XOR             |
| Hamming distance | XOR + bit count |

Two unique numbers     $\text{XOR} + x \ \& \ (-x)$

Number complement     $(\sim n) \ \& \ \text{mask}$

Generate subsets        Bitmask

$n \leq 20$                 Consider bitmasking